

Yohan Lee

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Research Interests

My research is driven by a fundamental concern: as AI systems grow increasingly powerful, the risks of concentrated control and deliberate misuse demand proactive, principled countermeasures. I focus on two interrelated challenges — (1) preventing the monopolization of AI by individuals or corporations, and (2) developing preemptive methodologies to defend against the malicious exploitation of AI. Ultimately, my goal is to establish robust frameworks that anticipate and mitigate the latent dangers of advanced AI before they materialize, ensuring that transformative AI technologies serve the broad public good rather than becoming instruments of harm.

Education

Sungkyunkwan University B.S. in Computer Science and Engineering	Feb. 2023 – Aug. 2027 (Expected) Suwon, Korea
Myongji University B.S. in Computer Engineering (transferred to SKKU)	Feb. 2021 – Dec. 2022

Publications

[P1] Yohan Lee*, Sujung Oh*, Gaeun Ko, and Sangmin Lee†. “Generate and Calibrate: Spectral Contrastive Flow Matching for Audio-Visual Sound Separation.” *Under review*.

*Equal contribution †Corresponding author

Research Experience

Pixel Lab, Sungkyunkwan University / Korea University <i>Undergraduate Researcher (Advisor: Prof. Sangmin Lee)</i>	Feb. 2025 – Present Seoul / Suwon, Korea
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- Designed a contrastive training objective (Spectral Contrastive Flow Matching) that prevents one generated output from leaking into another, using repulsive terms and hard negatives constructed by swapping frequency bands.
- Built an inference-time calibration step (Energy-Guided Calibration) that enforces a hard constraint — the observed input as an energy budget — on what the model may emit, correcting systematic over- and under-generation.
- Owned the work end to end as co-first author: problem formulation, objective design, PyTorch implementation, ablations, and evaluation on two public benchmarks (14.67 dB SDR on MUSIC).

V-Lab, Sungkyunkwan University <i>Undergraduate Researcher, URP (Advisor: Prof. Eunbyung Park)</i>	Jul. 2024 – Aug. 2024 Suwon, Korea
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- Built a diffusion-based (DDPM) pipeline for 3D model synthesis with text-driven texturing, improving texture consistency through loss and geometry adjustments.

AIoT Lab, Seoul National University <i>Undergraduate Researcher (Advisor: Prof. Hyung-Sin Kim)</i>	Dec. 2023 – Feb. 2024 Seoul, Korea
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- Studied robustness and out-of-distribution failure of vision models under sensor-induced distribution shift alongside the ImageNet-ES (CVPR 2024) team; worked on model compression for on-device deployment (Google ExploreCSR).

Honors & Awards

Grand Prize (1st place), Ambient AI Competition Seoul National University GSDS	Aug. 2023
Academic Excellence Scholarship Myongji University	2021, 2022

Selected Project

Sign Language Interpreter for Emergency Situations PyTorch, MediaPipe, Flutter, Flask	Aug. 2023
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- Curated a 70-word medical sign-language dataset and trained LSTM/GRU sequence classifiers over 30-frame MediaPipe keypoint windows; deployed as a Flutter tablet client on a Flask inference server. [Demo]

Technical Skills

Programming: Python, C/C++ | **Frameworks:** PyTorch, JAX/Flax, OpenCV
Tools: Git, Linux, Docker, WandB, $\text{\LaTeX}$ | **Languages:** Korean (native), English (professional)